



WORK EXPERIENCES

Data Engineer / Toyota Connected Europe / Dec 2024 – Now

- **AI-Native Engineering:** Standardised codebase architecture to be "LLM-readable," optimising for semantic clarity and context window efficiency; reduced token consumption by restructuring legacy patterns into modular, AI-ready components.
- **Agentic Automation:** Spearheading the internal transition toward autonomous operations by building prototypes for task-automation agents designed to handle repetitive engineering toil (e.g., automated PR analysis and documentation updates).
- **Semantic Data Quality:** Developed a custom framework for validating telematics data using semantic reasoning, moving beyond deterministic schema checks to ensure data integrity for downstream ML models.
- **Internal Tooling & DX:** Engineered a suite of internal developer tools and managed telemetry pipeline workflows using Airflow, prioritising modular design and auditability to ensure "production-grade" reliability across vehicle data streams.

Data Engineer / ASDA / Oct 2023 – Nov 2024

- **Engineering Excellence:** Developed and maintained production-grade internal Python packages to standardise ETL workflows, improving deployment velocity and code reusability across the data org.
- **Identity Resolution:** Engineered end-to-end Spark pipelines for customer data unification, implementing automated data quality frameworks and deterministic validation suites to ensure "gold-standard" record accuracy.

Consultant in Data Engineering & Architecture / Ekimetrics / Dec 2021 – Aug 2023

- **LLM & RAG Prototyping:** Built early-stage Proof-of-Concepts using LangChain and Vector Databases, focusing on document retrieval and context injection for enterprise-specific use cases.
- **Global Scalability:** Architected and deployed multi-region data lakehouses on Azure Databricks for a global automotive lead, synchronising data across 17 countries while enforcing strict GDPR and governance protocols.
- **Serverless Orchestration:** Designed AWS-native ingestion engines using Lambda and containerised microservices to handle high-volume event streams with integrated automated quality checks.
- **MLOps:** Led the development of end-to-end ML pipelines on Databricks, processing multi-million record datasets for advanced segmentation and predictive modeling.
- **Rapid-Response Infrastructure:** Engineered a lightweight data pipeline using Prefect for orchestration and Terraform for Infrastructure as Code (IaC), developing modular templates that standardised internal deployment patterns and accelerated project delivery

Engr Project Specialist (Innovation Lab) / Hewlett Packard Enterprise / Jun 2020 – Dec 2021

- **Enterprise Lakehouse Consulting:** Provided technical architecture guidance on HPE Ezmeral, demonstrating cloud-native data orchestration with Kubernetes and hybrid-cloud AI/ML capabilities.
- **Distributed Systems:** Developed data pipelines utilising the Hadoop ecosystem (Hive, Kafka, Spark) to support large-scale enterprise analytics.

EDUCATION

AI Performance Engineering / Nebius Academy / 2026 – Now

- ✓ Agents, LLM architecture, MLOps, Performance engineering, LLM post-training

MSc in Big Data Technology / The Hong Kong University of Science and Technology / 2020 – 2022

- ✓ Distributed Computing, Parallel Computing, Natural Language Processing, Machine Learning, Time Series Analysis, Data Mining

BEng(Hons) in Electronic and Information Engineering / The Hong Kong Polytechnic University / 2015 – 2020

- ✓ Machine Learning and IoT, Circuit Analysis and Logic Design, Embedded System Programming, Microcontroller System and Interface Design